Linux Distribution Benchmark Comparison

A Quick Taste Test of Linux Flavors

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***Abstract*—This report aims to highlight the strengths and weaknesses of some of the most commonly used Linux Distributions. In order to do this, a series of benchmarking tests were run on the distributions: Debian, Ubuntu, and Pop OS. The results of these tests were compared allowing any user to determine the best distribution for their needs.**

***Keywords*—Linux; distribution; performance; benchmarking; compiling; scheduling; Debian; Ubuntu; Pop OS; kernel**

I. Introduction

Linux has always had a special place in the world of operating systems. Although always overshadowed by its competitors: Windows and Mac; its ever growing following of loyal enthusiasts are adamant about Linux's superiority. In the desktop space, Linux has never cracked 4.5% of the market share [1], but it has been on an almost linear rise since its inception in 1991, slowly chipping away at the OS superpowers of the world.

Linux at its most basic definition is merely the kernel on which thousands of different distributions (referred to as "distros" from this point on) have been built. Because of this, one person's linux OS could look vastly different than another's. With a daily increase in linux users, many people are forced to blindly select the linux distro that they will install and defend for the rest of their life.

While guides exist for which linux distro to choose, they are predominantly geared towards the Graphical User Interface (GUI) of the distros and not the performance, which I believe to be of far greater performance in a primarily terminal based OS. It is my hope that this report will aid a budding Linux user in the choosing of their first distro.

With this goal in mind, the performance of three of the most common Linux distros have been tested and their results recorded.

II. What is a Linux Distribution

A. General Overview

A Linux distro builds on top of the Linux kernel combining libraries, utilities, and applications into a user friendly OS. While the GUI of each distro differs greatly, the average linux user should look first towards their terminal for getting the vast majority of tasks done. With this in mind, the most important differentiation of distros is their handling of processes and the impact this has on certain everyday tasks.

B. Debian

Debian is one of the first and most influential versions of linux having been released less than 3 years after the release of the first Linux kernel in 1991 [2].

Debian adopts the ondemand Central Processing Unit (CPU) frequency scaling tactic which dynamically boosts CPU frequency based upon the current load. This tactic helps to balance the system's power to performance ratio but induces a micro-pause at the introduction of heavy loads due to its periodic, time based trigger responsible for checking the CPU load at set intervals.

Debian also uses the mq-deadline I/O scheduler which enforces the standard First In First Out (FIFO) methodology, but separates its read and write operations into separate queues and heavily favors the read operations [4].

C. Ubuntu

Ubuntu is known and generally recommended for its wide application (and gaming) support. This distro - released in 2004 [3] - is younger and unapologetically derivative of Debian, however it has become more widely adopted and supported than its older brother.

Ubuntu runs a CPU governor called schedutil. Like Debian's ondemand, it adjusts the CPU frequency based on its current load; however it checks the load after every process change minimizing any latency that would result from polling the processes at a certain rate.

Ubuntu defaults to budget fair queuing (bfq) which is based on proportional-share scheduling. This process divides bandwidth into budgets of CPU time which are distributed through the queue proportional to weights. This allows some resources to be reserved and saved for critical tasks. Ubuntu prefers to spend this on user interaction requests allowing for quicker response times and immediate visual/audio outputs [4].

D. Pop OS

Pop OS is a distro copied straight from Ubuntu with some minor GUI changes and better integration with Graphics Processing Units (GPU). Created in 2017, the OS is fairly new however being built straight off the back of Ubuntu's work it fits in just fine with the older distros.

Having been copied from Ubuntu, Pop OS utilizes the same schedutil CPU scaler and bfq I/O scheduler. One differentiation of Pop OS is that it comes with fewer pre-installed programs minimizing bloatware freeing up more storage and memory as well as minimizing unwanted background tasks.

III. Why Performance Benchmarking Matters

While a computer can be a form of entertainment through video games, video content, or social media, it is first and foremost a tool, and finding the right OS for the primary applications needed will save the user time and energy in the future. In order to ascertain the ideal distro for a user's main tasks, the systems will need to be tested and compared for a variety of use cases.

A. CPU Compiling

The first test case is to compare each distro's time efficiency in file compilation. This is an important test for users who spend a lot of time programming or compiling files on their device. There are parts of development that require repetitive editing and compiling until a hidden bug is found and quicker compilation times can drastically improve the time required to do so.

B. CPU Single Threaded Tasks

When running programs or even interacting with the computer in general it is also very important to have minimal delay in communication between the OS and the CPU. Testing the latency in single threaded tasks helps to measure how much time the system spends doing nothing between the OS assigning a task and the CPU starting that task. This is very important for interactive applications that benefit from quick response.

C. Writing To The Disk

For workloads that require downloading or editing large amounts of data, quick write time is crucial for completing tasks in a time effective manner. Benchmarking the speed at which each OS can write to the disk enables the user to pick the distro that will optimize efficiency in every download, file edit, and install.

D. Reading From The Disk

On the other side of that same coin, reading from the disk is also crucial in data driven work where time waiting for the OS to open a file is time wasted. Every workload requires file reading of some kind, and comparing the read speeds of each OS allows the user to pick the distro that will ultimately save them time.

IV. Benchmarking Results

Four benchmarking tests were run three times on each distro for data validation. These distros: Debian, Ubuntu, and Pop OS were run through the same VM with identical specs and each set to balanced mode.

Below are the benchmarking tests in full detail and the results for each are provided with a few helpful notes on the results. No single conclusion is shared, however, because the decision can be based solely on the user's specific unique set of needs.

A. CPU Wait Time

In order to accurately replicate real world CPU compilation into one standardized test, a makefile was created that required the system to clone, build, and clean its own kernel. When the command 'time make -j$(nproc)' was run, the OS would make the file on all 6 threads and output the real time, system time, and user time taken to complete the compilation.

By subtracting the system and user time from the real time the wait time can be obtained. The wait time is how long the process spent not running. This is largely due to the process being blocked on the I/O scheduler. It is pretty clear from the table below that Debian's I/O handling is superior to the others in CPU compilation tasks and Pop OS lags far behind.

TABLE I.

CPU Compile	**Distros**		
	Debian	***Ubuntu***	***Pop OS***
Wait Time (s)	39.72	56.21	86.33

Fig. 1. Average CPU Compiler Wait Time in Seconds

B. Latency In Single Threaded Tasks

To measure the latency between the OS and CPU, sysbench's prime number calculation workload was run with the command "sysbench cpu --cpu-max-prime=20000 run." This made a single thread on the CPU run through every integer 1-20000 and check for primality. Instead of focusing on the total time taken or CPU speed statistics, the latency profile provided all the information needed to determine the delay between the OS and CPU.

The numbers in the table below indicate that the differences in latency were negligible, at most 2.8 milliseconds (ms) over the course of a 10 second test which is well within the bounds of reasonable error.

TABLE II.

CPU Single Thread	**Distros**		
	Debian	***Ubuntu***	***Pop OS***
Total Latency (s)	9.9913	9.9931	9.9903

Fig. 2. Average Cumulative Latency in CPU Single Threaded Task

C. Disk Write Speed

To check the disk write speed of each OS, the flexible I/O tester (fio) was used in order to benchmark the speeds on-device without any networking bottlenecks or variability. The command "fio --name=seqwrite --size=1G --bs=1M --rw=write --direct=1" allowed for a 1 gibibyte (GiB) file to be written to in chunks of 1 mebibyte (MiB) with the keyword direct=1 which allowed the program to write straight to the disk rather than the OS's cache.

Pop OS outputted some incredible speeds that were first thought to be flukes, but it sustained the speeds test after test. There is no obvious differentiating factor compared to the other two distros, but Pop OS writes to the disk faster than the other two combined.

TABLE III.

Sequential I/O Write	**Distros**		
	Debian	***Ubuntu***	***Pop OS***
Write Speed (MB/s)	78.10	80.00	203.33

Fig. 3. Average Write Speed in Megabytes per Second

D. Disk Read Speed

To measure disk read speed is a very similar task to measuring write speed. Fio was utilized once again by running “fio --name=randread --size=1G --bs=4K --rw=randread --iodepth=16” which forced the OS to read random 4 kibibyte (KiB) chunks from a 1 GiB file with 16 read requests queued to test how each distro handles multiple outstanding reads.

Debian won this benchmarking test by a relatively modest sum of 0.74 megabytes per second (MB/s) over Ubuntu, and Pop OS is brought back down to earth with a last place with 21.33 MB/s.

TABLE IV.

Random I/O Read	**Distros**		
	Debian	***Ubuntu***	***Pop OS***
Read Speed (MB/s)	23.47	22.73	21.33

Fig. 4. Average Read Speed in Megabytes per Second

E. Closing Thoughts

Each distro brings its own pros and cons to the table and it is up to the user to determine what is best for him or her.

Debian excelled at CPU compilation and had the best disk read speed, but it also had the worst write speed of the group. Ubuntu remained squarely in the middle in every relevant test with no upside against its counterparts but also no downsides. Finally, Pop OS blew away any competition with its write speed, but was the slowest in CPU compilation and disk read speed.

There is no perfect distro for everyone, but the ability to find one that suits a specific user’s needs is what differentiates Linux from other OSs.

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